

DAT 301 Project 1

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Introduction

- Happiness holds significance globally, and the World Happiness Report offers insights into worldwide well-being dynamics.
- The data set encompasses happiness scores and associated factors for 137 countries.
- Question: What factors contribute to disparities in happiness levels across countries, and how can we address these disparities effectively?**

Loading the Data

```
happiness_data <- read.csv("WHR2023.csv")

# Define original and new column names
original_names <- c("Country_name", "Ladder_score", "Standard_error", "Upper_whisker",
                    "Lower_whisker", "Logged_GDP_per_Capita", "Social_support",
                    "Healthy_life_expectancy", "Freedom_to_make_life_choices",
                    "Generosity", "Perceptions_of_corruption", "Ladder_score_in_Dystopia",
                    "Explained_by_Log_GDP_per_capita", "Explained_by_Social_support",
                    "Explained_by_Healthy_life_expectancy",
                    "Explained_by_Freedom_to_make_life_choices", "Explained_by_Generosity",
                    "Explained_by_Perceptions_of_corruption", "Dystopia_residual")

new_names <- c("Country", "Score", "Error", "Upper_Whisker", "Lower_Whisker",
               "GDP_per_Capita", "Social_Support", "Life_Expectancy",
               "Freedom", "Generosity", "Corruption_Perception",
               "Dystopia_Score", "Explained_GDP",
               "Explained_Social_Support", "Explained_Life_Expectancy",
               "Explained_Freedom", "Explained_Generosity",
               "Explained_Corruption_Perception", "Dystopia_Residual")

names(happiness_data) <- new_names
```

We loaded the data set from an Excel file and adjusted the column names for usability.

Top 10 Happiest Countries

```
top_10_happiest_countries <- happiness_data[order(happiness_data$Score, decreasing = TRUE), ][1:10, c("Country",
"Score")]

# Calculate mean GDP per capita, social support, and life expectancy for the top 10 happiest countries
mean_gdp <- sapply(top_10_happiest_countries$Country, function(country) {
  mean(happiness_data$GDP_per_Capita[happiness_data$Country == country])
})

mean_social_support <- sapply(top_10_happiest_countries$Country, function(country) {
  mean(happiness_data$Social_Support[happiness_data$Country == country])
})

mean_life_expectancy <- sapply(top_10_happiest_countries$Country, function(country) {
  mean(happiness_data$Life_Expectancy[happiness_data$Country == country])
})

# Add mean GDP per capita, social support, and life expectancy to the table
top_10_happiest_countries$Mean_GDP_per_Capita <- mean_gdp
top_10_happiest_countries$Mean_Social_Support <- mean_social_support
top_10_happiest_countries$Mean_Life_Expectancy <- mean_life_expectancy

# Print the updated table
print("Top 10 Happiest Countries with Mean GDP per Capita, Social Support, and Life Expectancy:")
```

```
## [1] "Top 10 Happiest Countries with Mean GDP per Capita, Social Support, and Life Expectancy:"
```

```
print(top_10_happiest_countries)

##      Country Score Mean_GDP_per_Capita Mean_Social_Support
## 1    Finland 7.804          10.792           0.969
## 2    Denmark 7.586          10.962           0.954
## 3    Iceland 7.530          10.896           0.983
## 4     Israel 7.473          10.639           0.943
## 5 Netherlands 7.403          10.942           0.930
## 6    Sweden 7.395          10.883           0.939
## 7    Norway 7.315          11.088           0.943
## 8 Switzerland 7.240          11.164           0.920
## 9    Luxembourg 7.228          11.660           0.879
## 10 New Zealand 7.123          10.662           0.952
##      Mean_Life_Expectancy
## 1              71.150
## 2              71.250
## 3              72.050
## 4              72.697
## 5              71.550
## 6              72.150
## 7              71.500
## 8              72.900
## 9              71.675
## 10             70.350
```

We identify and analyze the top 10 happiest countries, computing mean factors to understand their association with happiness.

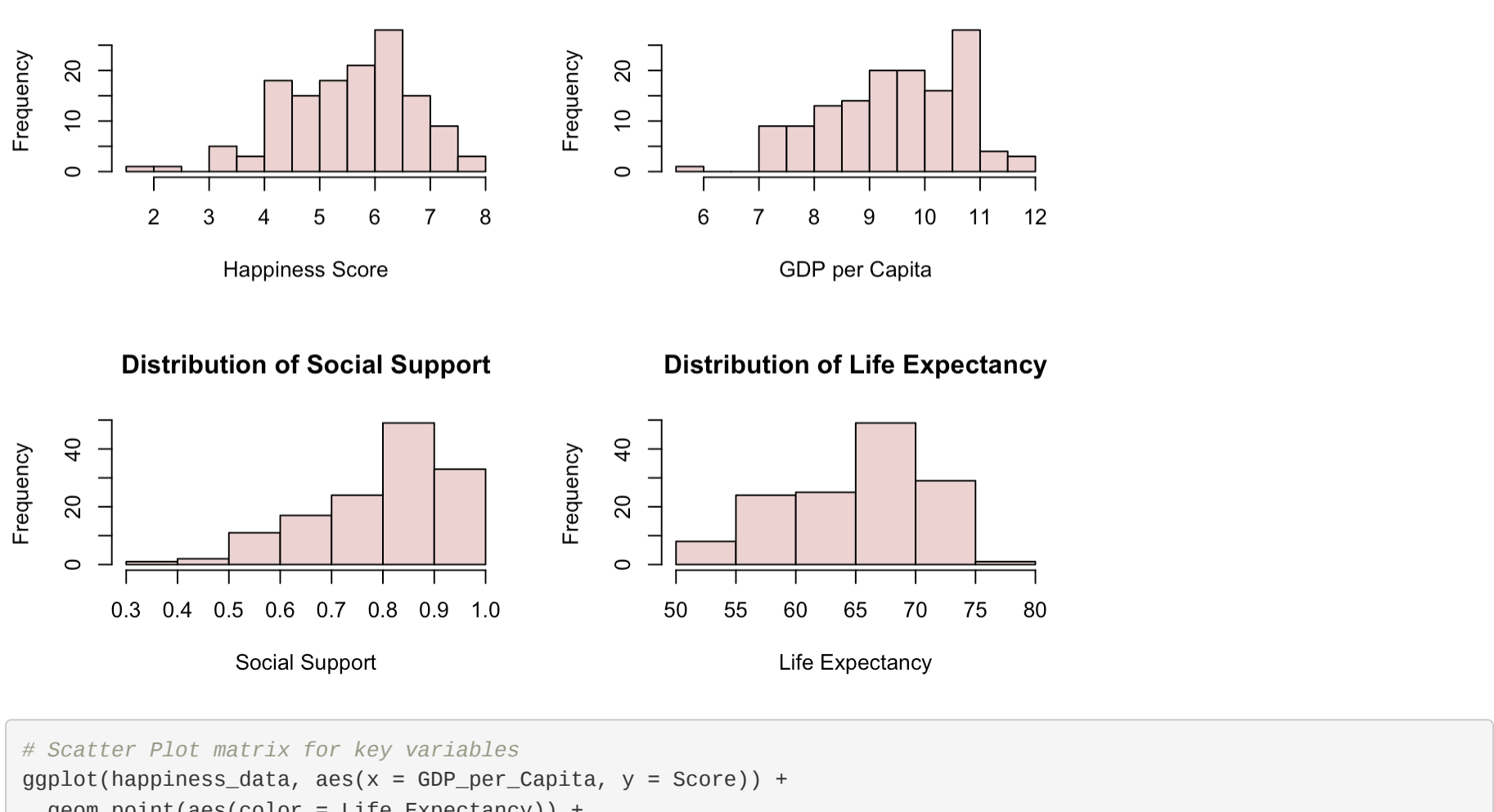
Exploring Well-being: Analysis of Key Variables

```
library(ggplot2)

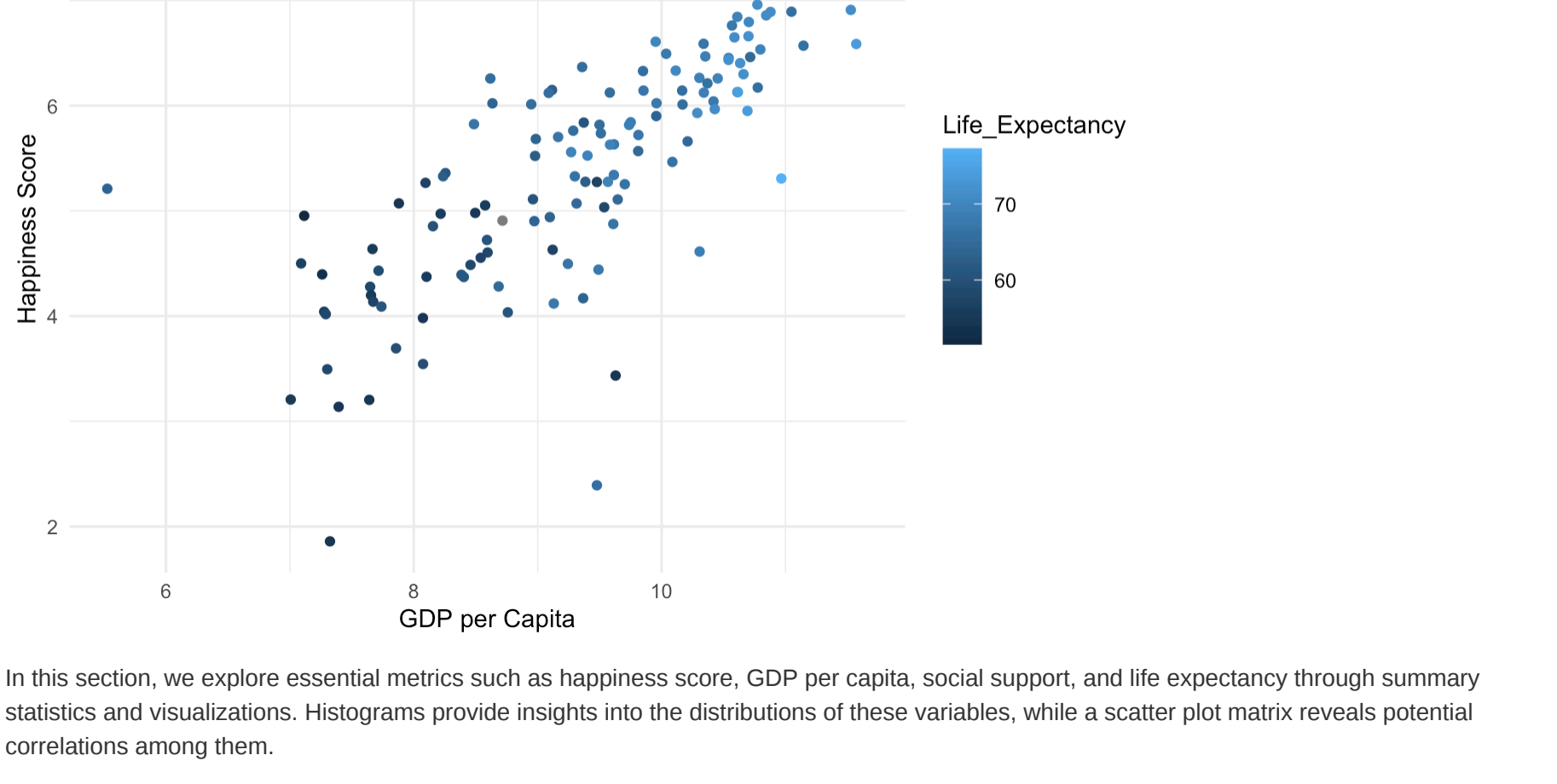
# Summary statistics for key variables
summary(happiness_data[, c("Score", "GDP_per_Capita", "Social_Support", "Life_Expectancy")])

##      Score      GDP_per_Capita      Social_Support      Life_Expectancy
##  Min.   :1.859   Min.   : 5.527   Min.   :0.3410   Min.   :51.53
##  1st Qu.:4.724   1st Qu.: 8.591   1st Qu.:0.7220   1st Qu.:60.65
##  Median :5.684   Median : 9.567   Median :0.8270   Median :65.84
##  Mean   :5.540   Mean   : 9.450   Mean   :0.7991   Mean   :64.97
##  3rd Qu.:6.334   3rd Qu.:10.540   3rd Qu.:0.8960   3rd Qu.:69.41
##  Max.   :7.804   Max.   :11.660   Max.   :0.9830   Max.   :77.28
##                                     NA's   :1

# Histograms for key variables
par(mfrow = c(2, 2)) # Set up a 2x2 grid for plots
hist(happiness_data$Score, main = "Distribution of Happiness Scores", xlab = "Happiness Score", col = "mistyrose2")
hist(happiness_data$GDP_per_Capita, main = "Distribution of GDP per Capita", xlab = "GDP per Capita", col = "mistyrose2")
hist(happiness_data$Social_Support, main = "Distribution of Social Support", xlab = "Social Support", col = "mistyrose2")
hist(happiness_data$Life_Expectancy, main = "Distribution of Life Expectancy", xlab = "Life Expectancy", col = "mistyrose2")
```



```
# Scatter Plot matrix for key variables
ggplot(happiness_data, aes(x = GDP_per_Capita, y = Score)) +
  geom_point(aes(color = Life_Expectancy)) +
  labs(title = expression(paste(bold("Scatter Plot: GDP, Happiness, and Life Expectancy Correlations"))),
        x = "GDP per Capita",
        y = "Happiness Score") +
  theme_minimal()
```



In this section, we explore essential metrics such as happiness score, GDP per capita, social support, and life expectancy through summary statistics and visualizations. Histograms provide insights into the distributions of these variables, while a scatter plot matrix reveals potential correlations among them.

Correlation Analysis: Factors Associated with Happiness Scores

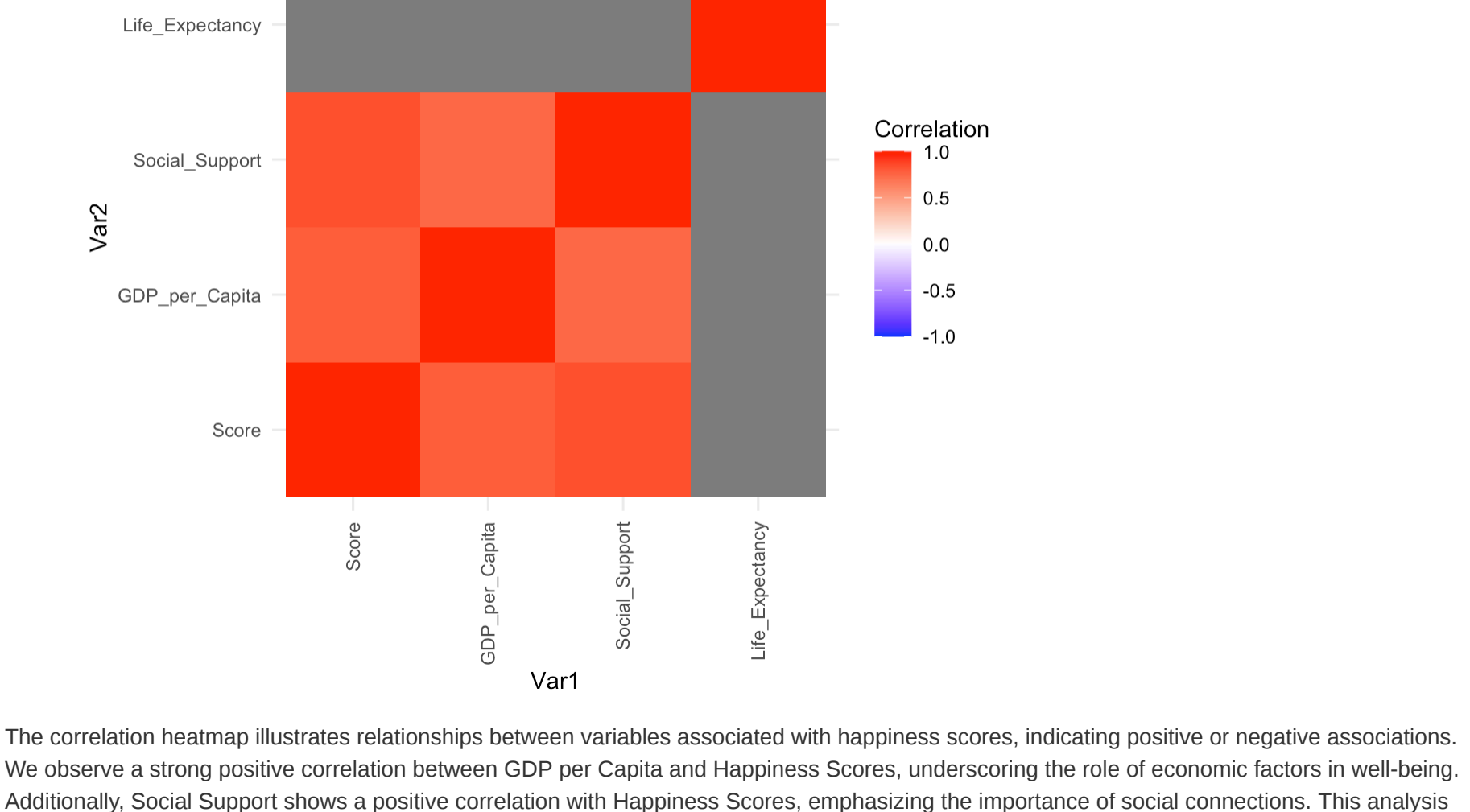
```
correlation_matrix <- cor(happiness_data[,c("Score", "GDP_per_Capita", "Social_Support", "Life_Expectancy")])

print(correlation_matrix)

##      Score      GDP_per_Capita      Social_Support      Life_Expectancy
##  Score      1.0000000      0.7843673      0.8345317      NA
##  GDP_per_Capita 0.7843673      1.0000000      0.7380688      NA
##  Social_Support 0.8345317      0.7380688      1.0000000      NA
##  Life_Expectancy NA      NA      NA      NA      1

# Visualize correlation matrix using heatmap
library(ggplot2)
library(reshape2)

melted_correlation <- melt(correlation_matrix)
ggplot(data = melted_correlation, aes(x = Var1, y = Var2, fill = value)) +
  geom_tile() +
  scale_fill_gradient2(low = "blue", high = "red", mid = "white",
                       midpoint = 0, limit = c(-1,1), space = "Lab",
                       name="Correlation") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1)) +
  coord_fixed()
```



The correlation heatmap illustrates relationships between variables associated with happiness scores, indicating positive or negative associations. We observe a strong positive correlation between GDP per Capita and Happiness Scores, underscoring the role of economic factors in well-being. Additionally, Social Support shows a positive correlation with Happiness Scores, emphasizing the importance of social connections. This analysis sheds light on factors influencing happiness disparities.

Regional Disparities in Happiness

```
library(dplyr)

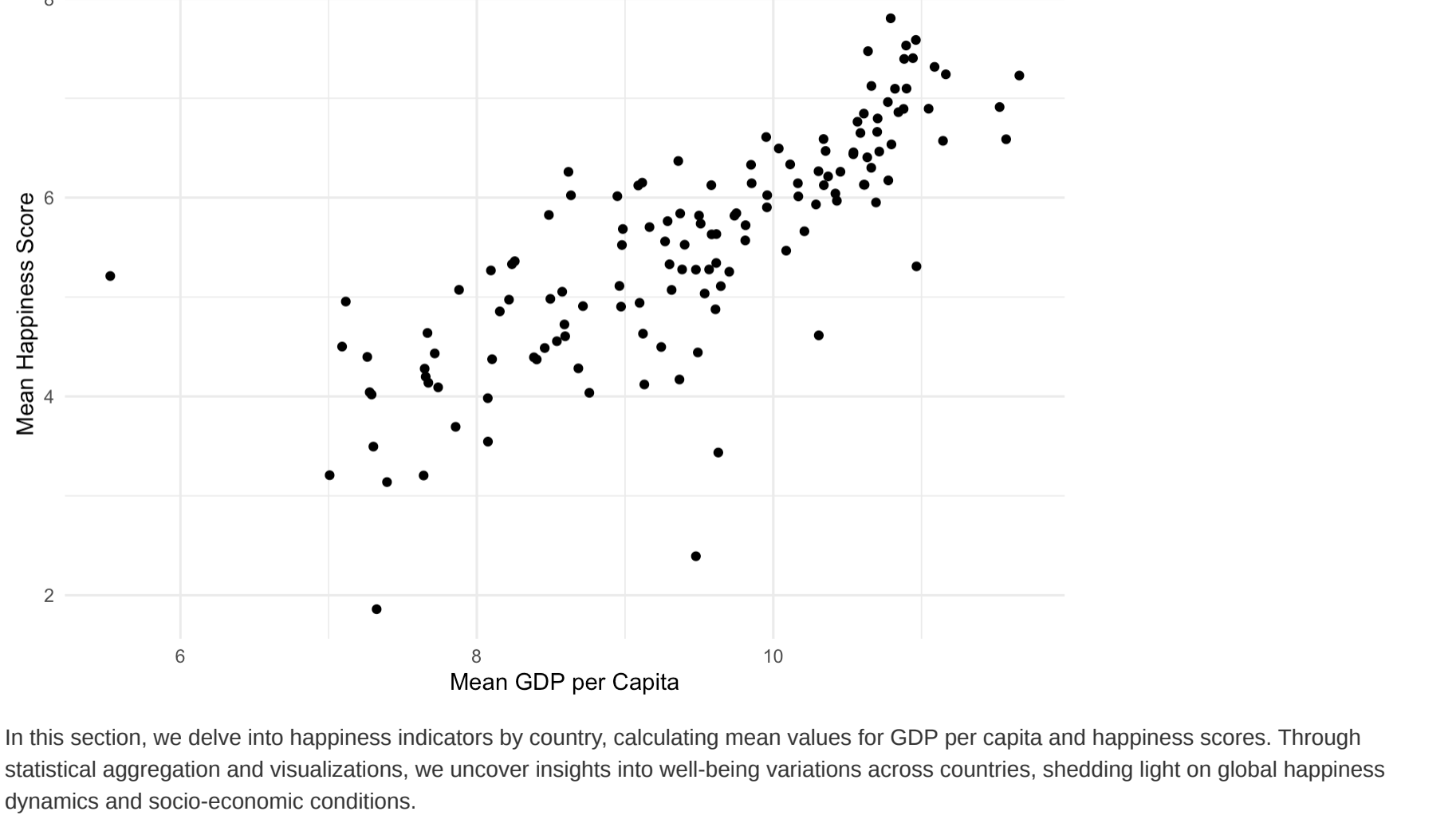
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

country_summary <- happiness_data %>%
  group_by(Country) %>%
  summarise(
    Mean_Happiness_Score = mean(Score),
    Mean_GDP_per_Capita = mean(GDP_per_Capita),
    Mean_Social_Support = mean(Social_Support),
    Mean_Life_Expectancy = mean(Life_Expectancy)
  )

# Visualize happiness indicators among countries
# Scatter plot comparing GDP per capita and happiness scores by country
ggplot(country_summary, aes(x = Mean_GDP_per_Capita, y = Mean_Happiness_Score)) +
  geom_point() +
  labs(title = "GDP per Capita vs. Happiness Score by Country",
        x = "Mean GDP per Capita",
        y = "Mean Happiness Score") +
  theme_minimal()
```



In this section, we delve into happiness indicators by country, calculating mean values for GDP per capita and happiness scores. Through statistical aggregation and visualizations, we uncover insights into well-being variations across countries, shedding light on global happiness dynamics and socio-economic conditions.

Identifying Outliers and Anomalies

```
# Define outlier detection function
detect_outliers <- function(data, variable) {
  q1 <- quantile(data[[variable]], 0.25)
  q3 <- quantile(data[[variable]], 0.75)
  iqr <- q3 - q1
  lower_bound <- q1 - 1.5 * iqr
  upper_bound <- q3 + 1.5 * iqr
  outliers <- data[data[[variable]] < lower_bound | data[[variable]] > upper_bound, ]
  return(outliers)
}

# Detect outliers in happiness scores
happiness_outliers <- detect_outliers(happiness_data, "Score")

# Investigate countries with exceptionally high or low happiness scores
print("Countries with exceptionally high or low happiness scores:")

## [1] "Countries with exceptionally high or low happiness scores:"

print(happiness_outliers)

##      Country Score Error Upper_Whisker Lower_Whisker GDP_per_Capita
## 137 Afghanistan 1.859 0.633          1.923          1.795          7.324
##      Social_Support Life_Expectancy Freedom Generosity Corruption
## 137          0.341          54.712      0.382      -0.081          0.847
##      Dystopia_Score Explained_GDP Explained_Social_Support
## 137          1.778          0.645              0
##      Explained_Life_Expectancy Explained_Freedom Explained_Generosity
## 137          0.087          0.087              0          0.093
##      Explained_Corruption_Perception Dystopia_Residual
## 137          0.059          0.976
```

In this section, we aim to identify outliers or anomalies within the dataset that might significantly influence our analysis. By detecting countries with exceptionally high or low happiness scores relative to their economic and social factors, we can gain insights into potential outliers that warrant further investigation.

Conclusion

Our analysis of the World Happiness Report data across 137 countries reveals crucial insights into global well-being dynamics. Top-ranking nations like Finland and Denmark exhibit high happiness scores, often correlated with robust GDP per capita, social support, and life expectancy.

Key variables such as **happiness score**, **GDP per capita**, and **social support** exhibit meaningful correlations, emphasizing the significance of economic and social factors in happiness levels. Regional disparities, exemplified by outliers like Afghanistan, underscore the need for targeted interventions to address underlying socio-economic challenges.

In conclusion, prioritizing equitable economic growth, robust social support structures, and accessible healthcare systems can help governments and policymakers mitigate happiness disparities and foster greater well-being worldwide.

Possible Policy implications and Recommendations

- Inclusive Growth:** Prioritize inclusive economic policies to reduce income inequality and enhance overall well-being through education, job creation, and entrepreneurship.
- Social Support Networks:** Strengthen healthcare, social services, and community programs to build resilience and foster happiness amid adverse circumstances.
- Transparent Governance:** Establish transparent governance and anti-corruption measures to foster public trust and create environments conducive to happiness and prosperity.

Future Research Directions:

- Longitudinal Studies:** Conduct longitudinal studies to track changes in happiness indicators over time and understand the long-term impact of policy interventions and socio-economic trends.
- Cross-Cultural Analysis:** Explore cross-cultural variations in happiness determinants and examine how cultural factors influence subjective well-being across diverse populations.
- Technology and Well-being:** Investigate the role of technology in shaping happiness levels, including its impact on social interactions, mental health, and work-life balance.